

Flow of Control

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Selection:

1. `if condition:`
 `statement(s)`

2. `if condition:`
 `statement(s)`
 `else:`
 `statement(s)`

3. `if condition:`
 `statement(s)`
 `elif condition:`
 `statement(s)`
 `elif condition:`
 `statement(s)`
 :
 :
 `else:`
 `statement`

Nested:

4. `if condition:`
 `if condition:`
 :
 `elif condition:`
 `statement:`
 `if condition:`

Any level of nesting is required.
we can do.

`match (variable):`
 `case value:`
 `statements`
 `case value:`
 `statement(s)`
 :
 :
 `case _:`
 `statement(s)`

`variable == value`,
that only case will be
executed.

} when all cases fails then '-' case
will execute.

```
num1 = int(input('Enter first number: '))
num2 = int(input('Enter second number: '))
op = input('Enter operator (+,-,*,/): ')
```

```
match op:
```

```
    case '+':
        print(f'Sum = {num1 + num2}')
```

```
    case '-':
        print(f'Difference = {num1 - num2}')
```

```
    case '*':
        print(f'Product = {num1 * num2}')
```

```
    case '/':
        if num2 == 0:
```

↓ or F ↓ expression .

↑ formatted string ↓ place holder for data/variable.

```

case '/': formatted thing
    if num2 == 0:
        print('Division by 0 is not allowed')
    else:
        print(f'Quotient = {num1 / num2}')
case _:
    print('Wrong operator selected')

```

```

>>> price = 1234.5678
>>> print(f'Price: {price:.2f}')
Price: 1234.57
>>> print(f'Price: {price:.3f}')
Price: 1234.568
>>> print(f'Price: {price:,.3f}')
Price: 1,234.568

```

number of digits after decimal place.

comma as thousand separator

```

score = int(input('Enter your score: '))
print(f"Result: {'Pass' if score >= 50 else 'Fail'}")

```

Format	Example	Output
Decimal integer	f"{42:d}"	42
Leading zeros	f"{42:05d}"	00042
Leading spaces	f"{42: 5d}"	42
Fixed-point	f"{3.14159:.2f}"	3.14
Comma separator	f"{100000:,}"	1,000,000
Binary	f"{10:b}"	1010

For example:

```

score = int(input('Enter your score: '))
print(f"Result: {'Pass' if score >= 50 else 'Fail'}")
print(f"{score:03d}")
print(f"{score:3d}")
print(f"{1000000:,}")
print(f"{score:b}")

```

Output:

```

Enter your score: 10
Result: Fail
010
10
10,000,000
1010

```

Repetition :

```

print(1)
print(2)

```

```
print(3)
print(4)
```

while condition:
 Statement(s)

For example:
 I want to print values from 1 to 10.

```
start = 1
stop = 10
while start <= stop:
    print(start)
    start += 1
```

```
start = int(input('Enter the start value: '))
stop = int(input('Enter the stop value: '))
while start <= stop: ← when this condition is false then else block will execute.
    print(start)
    start += 1
    if start == stop: #if this condition is true then else part of while loop
will not work
        break
else: ← not compulsory, if required use it.
    print('Loop ends')
```